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Live App URL: https://one.swkpropertiesuae.com · ae.swkproperties.one
GitHub Repo: github.com/swkpropertiesuae/SWKONE

DOC-02 OF 04 · TECHNICAL ARCHITECTURE MANUAL · ANDROID, HYBRID AI, DATABASE & CLOUD RUNBOOK

SWK Properties LLC — SWK ONE Platform

www.swk.properties · www.swkpropertiesuae.com · www.swk.rent

DOCUMENT 02 · TECHNICAL ARCHITECTURE, NATIVE ANDROID, HYBRID AI & CLOUD MANUAL

5-Tier System Topology, Android Bridge, AI Gateway & Runbook

Complete engineering reference for maintaining, building, backing up, and deploying **SWK ONE** across Native Android (`ae.swkproperties.one`), Cloudflare Containers (`swk-one`), Cloudflare Workers AI + D1 (`deploy/one-ai`), Private Ollama, and PostgreSQL 16.

1. 5-Tier Cloud, Mobile & Hybrid AI Architecture

Tier 1: Native Android (`ae.swkproperties.one`) & PWA

Bundled WebView + Native Java Bridge (`SwkNativePlugin.java`) and Foreground GPS Service (`SwkDutyLocationService.java`). Provides hardware device ID enrollment, live GPS coordinates during active Field Duty, battery/network telemetry, biometric lock, and encrypted offline action queue (`swk_one_offline_queue_v1`).

Tier 2: Cloudflare Edge Worker & Container (`swk-one`)

Traffic to `one.swkpropertiesuae.com` (`deploy/cloudflare/wrangler.jsonc`) routes to the `SwkOs` Durable Object Container (`node:22-bookworm-slim` with headless Chromium, Poppler PDF utilities, and English + Arabic Tesseract OCR packs).

Tier 3: Hybrid Cloud AI Worker & Private Ollama Engine

Public-approved writing & media tasks route through `deploy/one-ai` (Cloudflare Worker + D1 quota ledger `deploy/one-ai/schema.sql`) with 10-seat `STAFF_SLOTS` isolation. Confidential Title Deeds, contracts, and legal reviews execute strictly on private local Ollama (`qwen3.5:4b` / `qwen3.5:9b`) and never fall back to cloud providers.

Tier 4 & 5: Fastify 5 Core (384 Routes) & Dual PostgreSQL/SQLite

High-throughput Node.js 22 server (`src/server/app.ts`) exposing 384 Zod-validated endpoints backed by `src/server/db.ts` and `src/server/pg-sync.ts` , synchronizing all 75+ tables between local SQLite WAL and managed Cloud PostgreSQL 16.

2. Security, Privacy & Governance Specifications

SECURITY / PRIVACY LAYER	IMPLEMENTATION FILE	ENGINEERING SPECIFICATION
Password & Session Crypto	`src/server/auth.ts`	Node.js `crypto.scrypt` with unique 16-byte salts, timing-safe comparison, RFC 6238 TOTP 2FA, and cross-origin Android session token headers (`x-swk-session`).

SECURITY / PRIVACY LAYER	IMPLEMENTATION FILE	ENGINEERING SPECIFICATION
Hard Staff Access Guard	`src/shared/swk-staff-access.ts` `SwkOneRestrictedStaffView.tsx`	Restricts full CRM/Inventory/Accounting/Admin & QR printing exclusively to Owner (Shabib Siraj) & Manager (Mohammad Haseeb); locks standard `@swk.property` accounts into the 4-Tab Staff Portal.
Zero Off-Duty GPS Guarantee	`SwkDutyLocationService.java` `src/web/mobile/native-bridge.ts`	Location coordinates are captured ONLY while Field Duty is explicitly active or during a viewing arrival check-in. Stopping duty immediately halts the Android foreground service and logs an entry in the Location Privacy Audit Log.
AI Data Classification Gate	`src/server/routes/one-ai.ts` `deploy/one-ai/src/worker.mjs`	Cloud AI requests require `classification: "public"` and `confirmPublic: true`, plus a valid numeric seat in `STAFF_SLOTS` (1–10). Legal document review (`/api/one/review`) is hard-locked to private Ollama.
Tamper-Evident Audit Chain	`src/server/audit.ts`	Every record mutation logs before/after JSON diffs chained via `SHA-256(prev_hash + payload)` to detect any database tampering.

3. Build, Android APK Packaging & Deployment Commands

```
# 1. Install dependencies, typecheck & build production Web + API bundles:
npm ci && npm run typecheck && npm run build

# 2. Run automated SWK ONE Android (63/63 checks), Mobile GPS & Cloud AI Bridge test suites:
npx tsx tests/swk-one-android.ts
npx tsx tests/mobile.ts
node --import tsx tests/one-cloud-bridge.ts && node --test deploy/one-ai/src/worker.test.mjs

# 3. Sync Web bundle into Native Android project & build APK (android/app/build/outputs/apk/debug/):
npm run android:sync && npm run android:apk

# 4. Regenerate Official Reception Attendance QR Poster & Handover PDFs:
node tools/generate-attendance-qr-and-guide.mjs
```